Midterm Cheatsheet — Lectures 1–2

Definitions and theorems that are easy to forget. Vectors of $\mathbb{R}^n$ are columns.

Definitions people forget

Spaces and maps

- **Subspace**: contains 0, closed under + and scalar multiplication.
- **Linear independence** of an infinite set: every *finite* subset is independent; span = finite linear combinations.
- **Matrix of** $T : V \to W$: $[T]_{\mathcal{C},\mathcal{B}}$ has j-th column $[T(b_j)]_{\mathcal{C}}$; then $[T(v)]_{\mathcal{C}} = [T]_{\mathcal{C},\mathcal{B}}[v]_{\mathcal{B}}$.
- **Similar**: $B = P^{-1}AP$ for an invertible P.
- **Group**: associative, identity, inverses; **abelian** if $ab = ba$. $\mathrm{GL}(n)$, $\mathrm{SL}(n) = \{\det = 1\}$, $\mathrm{O}(n) = \{Q^\top Q = I\}$, $\mathrm{SO}(n) = \mathrm{O}(n) \cap \mathrm{SL}(n)$.

Duality and quotients

- **Dual space** $V^* = \{\varphi : V \to \mathbb{F} \text{ linear}\}$. **Dual basis**: $b^i(b_j) = \delta_{ij}$; for $\mathbb{R}^n$ with $P = (\mathbf{b}_1 \cdots \mathbf{b}_n)$, $b^i = i$-th *row* of P^{-1}.
- **Dual map** $T^* : W^* \to V^*$, $T^*(\psi) = \psi \circ T$; its matrix in the dual bases is $A^\top$.
- **Annihilator** $U^\circ = \{\varphi \in V^* : \varphi|_U = 0\}$.
- **Double dual**: $\iota(v)(\varphi) = \varphi(v)$; $V \cong V^{**}$ if $\dim V < \infty$.
- **Coset** $v + U$; $v + U = w + U \iff v - w \in U$. **Quotient** V/U = set of cosets, $(v + U) + (w + U) = (v + w) + U$, $c(v + U) = cv + U$, zero vector U; quotient map $\pi(v) = v + U$.
- **Sum** $U + W$; **direct sum** $V = U \oplus W$: every $v = u + w$ *uniquely*.

Matrices

- **Schur complement** of A in $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$: $S = D - CA^{-1}B$.
- **Cofactor** $C_{ij} = (-1)^{i+j} \det A^{(ij)}$ ($A^{(ij)}$: delete row i, column j).
- **Characteristic polynomial** $p_A(\lambda) = \det(A - \lambda I)$. **Algebraic multiplicity**: multiplicity as a root of p_A. **Geometric**: $\dim E_\lambda = n - \operatorname{rank}(A - \lambda I)$.
- **Minimal polynomial** m_A: monic, smallest degree, $m_A(A) = 0$.
- **Circulant** $\operatorname{circ}(\mathbf{c})_{jk} = c_{j-k \bmod n}$, matrix of $\mathbf{x} \mapsto \mathbf{c} * \mathbf{x}$.

Tensors

- **Tensor product** $V \otimes W = \mathcal{F}_0(V \times W)/R$, R spanned by the bilinearity relations; $v \otimes w = \delta_{(v,w)} + R$. **Pure tensor**: $v \otimes w$.
- **Tensor rank**: least number of pure tensors with sum t (zero tensor: 0).
- **Contraction** $C : V \otimes V^* \to \mathbb{R}$, $C(v \otimes \varphi) = \varphi(v)$.
- **Type** (p, q): element of $V^{\otimes p} \otimes (V^*)^{\otimes q}$. Vector $(1, 0)$, functional $(0, 1)$, linear map $(1, 1)$, bilinear form $(0, 2)$.
- **Einstein convention**: repeated indices are summed; the others are free.

Inner products and norms

- **Inner product**: bilinear, symmetric, positive definite. **SPD**: $A = A^\top$, $\mathbf{x}^\top A\mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$.
- **Orthogonal complement** $U^\perp = \{v : \langle v, u\rangle = 0 \ \forall u \in U\}$.
- **Orthogonal projection**: $\pi_U(v) \in U$ and $v - \pi_U(v) \in U^\perp$.
- **Spectral norm** $\|A\|_2 = \max_{\mathbf{x}\neq\mathbf{0}} \|A\mathbf{x}\|/\|\mathbf{x}\|$; **Frobenius** $\|A\|_F^2 = \sum_{ij} A_{ij}^2 = \operatorname{tr}(A^\top A)$.
- **Pseudo-inverse** $A^+ = V\Sigma^+U^\top$, $(\Sigma^+)_{ii} = 1/\sigma_i$ for $i \le r$.
- **Condition number** $\kappa(A) = \sigma_1/\sigma_n$ (independent columns).
- **Graph Laplacian** $L = D - W$, $D = \operatorname{diag}(d_i)$, $d_i = \sum_j w_{ij}$.

Theorems people forget

Dimension counting

- **Rank–nullity**: $\dim\ker T + \dim\operatorname{im} T = \dim V$. $\operatorname{rank} A = \dim\operatorname{col} A = \dim\operatorname{row} A$.
- **First isomorphism theorem**: $V/\ker T \cong \operatorname{im} T$ via $v + \ker T \mapsto T(v)$.
- $\dim V/U = \dim V - \dim U$; $(V/U)^* \cong U^\circ$; in $\mathbb{R}^n$: $\dim U + \dim U^\circ = n$.
- $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$. Second iso: $(U + W)/W \cong U/(U \cap W)$.
- **Fredholm**: $A\mathbf{x} = \mathbf{b}$ is solvable $\iff$ $\mathbf{y}\mathbf{b} = 0$ for every row $\mathbf{y}$ with $\mathbf{y}A = \mathbf{0}$.

Determinants, traces, inverses

- $\det(AB) = \det A \det B$, $\det A^\top = \det A$, $\det(cA) = c^n \det A$, $\det A^{-1} = 1/\det A$. Row swap: $\times(-1)$; scale a row by c: $\times c$; add a multiple of a row: unchanged.
- $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ (also for rectangular A, B); $\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A$.
- $A^{-1} = \frac{1}{\det A} C^\top$ (cofactor matrix C). For 2×2: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- $\det\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D$; $\det M = \det A \det S$ (Schur), and $M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{pmatrix}$.
- **Woodbury**: $(A+UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1}+VA^{-1}U)^{-1}VA^{-1}$.
- **Sherman–Morrison**: $(A + \mathbf{u}\mathbf{v}^\top)^{-1} = A^{-1} - \dfrac{A^{-1}\mathbf{u}\mathbf{v}^\top A^{-1}}{1 + \mathbf{v}^\top A^{-1}\mathbf{u}}$.
- **Determinant lemma**: $\det(I_n + UV) = \det(I_k + VU)$; $\det(A + \mathbf{u}\mathbf{v}^\top) = (1 + \mathbf{v}^\top A^{-1}\mathbf{u}) \det A$.
- **LU**: $PA = LU$ for invertible A (L unit lower, U upper triangular).

Eigenvalues

- $\det A = \prod \lambda_i$, $\operatorname{tr} A = \sum \lambda_i$ (complex, with multiplicity). 2×2: $p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det A$.
- Eigenvectors for distinct eigenvalues are independent. $1 \le$ geometric $\le$ algebraic.
- Diagonalizable $\iff$ n independent eigenvectors $\iff$ p_A has n real roots and geometric = algebraic for all λ $\iff$ m_A = product of *distinct* real linear factors.
- **Cayley–Hamilton**: $p_A(A) = 0$. m_A divides every q with $q(A) = 0$; the roots of m_A are the eigenvalues.
- $A = PDP^{-1} \Rightarrow A^k = PD^kP^{-1}$. A matrix commutes with the cyclic shift $\iff$ it is circulant; the DFT diagonalizes circulants.

Tensors

- **Universal property**: every bilinear $\beta : V \times W \to U$ gives exactly one linear $\tilde\beta : V \otimes W \to U$ with $\tilde\beta(v \otimes w) = \beta(v, w)$.
- $\{b_i \otimes c_j\}$ is a basis; $\dim(V \otimes W) = \dim V \dim W$. $\mathbb{R}^m \otimes \mathbb{R}^n \cong \mathbb{R}^{m\times n}$, $\mathbf{u} \otimes \mathbf{v} \mapsto \mathbf{u}\mathbf{v}^\top$; tensor rank = matrix rank.
- $\operatorname{Hom}(V, W) \cong W \otimes V^*$, $w \otimes \varphi \mapsto (v \mapsto \varphi(v)w)$; multilinear forms $\leftrightarrow V_1^* \otimes \cdots \otimes V_k^*$; tr = contraction.
- **Change of basis** $b'_j = \sum_i P_{ij} b_i$: vectors $\mathbf{x}' = P^{-1}\mathbf{x}$; linear maps $A' = P^{-1}AP$; bilinear forms $A' = P^\top AP$; $V \otimes V$: $A' = P^{-1}A(P^{-1})^\top$. Each V-index transforms with P^{-1}, each V^*-index with P.

Inner product spaces

- **Cauchy–Schwarz** $|\langle x, y\rangle| \le \|x\|\|y\|$, equality $\iff$ dependent. **Parallelogram**: $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$. **Polarization**: $\langle x, y\rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2)$.
- $\|\mathbf{x}\|_\infty \le \|\mathbf{x}\|_2 \le \|\mathbf{x}\|_1 \le n\|\mathbf{x}\|_\infty$.
- **Gram–Schmidt**: $u_j = v_j - \sum_{i<j}\langle v_j, e_i\rangle e_i$, $e_j = u_j/\|u_j\|$. **QR**: $A = QR$.
- ONB: $v = \sum_i \langle v, b_i\rangle b_i$, $\|v\|^2 = \sum_i \langle v, b_i\rangle^2$ (Parseval).
- $V = U \oplus U^\perp$, $(U^\perp)^\perp = U$; $\operatorname{null} A = (\operatorname{row} A)^\perp$, $\operatorname{null} A^\top = (\operatorname{col} A)^\perp$.
- **Projections**: onto a line $P = \frac{\mathbf{b}\mathbf{b}^\top}{\mathbf{b}^\top\mathbf{b}}$; onto $\operatorname{col} B$: $P = B(B^\top B)^{-1}B^\top$; $P^\top = P = P^2$. $\pi_U(v)$ is the closest point of U to v. Distance to $\{\mathbf{w}^\top\mathbf{z} = c\}$: $\frac{|\mathbf{w}^\top\mathbf{x} - c|}{\|\mathbf{w}\|}$.
- **Least squares**: $A^\top A\mathbf{w}^* = A^\top\mathbf{y}$. All minimizers: $A^+\mathbf{y} + \operatorname{null}(A)$; $A^+\mathbf{y}$ has the least norm.
- SPD $\iff$ all eigenvalues > 0 $\iff$ all leading minors > 0 (Sylvester) $\iff$ $A = LL^\top$, L lower triangular with positive diagonal (Cholesky).

Spectral theorem and SVD

- Symmetric $A = Q\Lambda Q^\top$, Q orthogonal, real eigenvalues. $A^{1/2} = Q\Lambda^{1/2}Q^\top$.
- **Rayleigh**: $\lambda_n \le \frac{\mathbf{x}^\top A\mathbf{x}}{\mathbf{x}^\top\mathbf{x}} \le \lambda_1$. **Ky Fan**: $\operatorname{tr}(B^\top AB) \le \lambda_1 + \cdots + \lambda_k$ for $B^\top B = I_k$. **Courant–Fischer**: $\lambda_k = \max_{\dim U = k} \min_{\mathbf{x}\in U\setminus 0} R_A(\mathbf{x})$. **Weyl** ($A$, E symmetric, eigenvalues of E in $[-\varepsilon, \varepsilon]$): $|\lambda_k(A + E) - \lambda_k(A)| \le \varepsilon$.
- **SVD** $A = U\Sigma V^\top = \sum_{i\le r} \sigma_i \mathbf{u}_i\mathbf{v}_i^\top$; σ_i^2 = eigenvalues of $A^\top A$;

$\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$. $\operatorname{col} A = \operatorname{span}\{\mathbf{u}_{1..r}\}$, $\operatorname{null} A = \operatorname{span}\{\mathbf{v}_{r+1..n}\}$.

- $\|A\|_2 = \sigma_1$, $\|A\|_F^2 = \sum \sigma_i^2$. **Eckart–Young**: $A_k = \sum_{i \le k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ is best of rank $\le k$; $\|A - A_k\|_2 = \sigma_{k+1}$, $\|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$.
- $\frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \le \kappa(A) \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|}$; $\kappa(A^\top A) = \kappa(A)^2$.
- **PCA** (centered data, $S = \frac{1}{N} X^\top X$): best k-dim subspace $\operatorname{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_k\}$, mean error $\lambda_{k+1} + \cdots + \lambda_d$.
- **Laplacian**: $\mathbf{x}^\top L \mathbf{x} = \sum_{i<j} w_{ij}(x_i - x_j)^2$; multiplicity of eigenvalue 0 = number of connected components.